

Business Rules & Analytical Definitions

1. Purpose

Define the rules and analytical measures used to identify delivery delays, compare customer satisfaction, and analyse delivery performance by customer state.

2. Business Problem

The e-commerce operations team lacks clear visibility into where delivery delays are occurring and whether delayed orders are associated with lower customer satisfaction. This makes it difficult to identify which areas should be prioritised for improvement.

3. Business Rules

ID	Rule Definition	Purpose
BR-01	Include only orders where order_status = 'delivered'.	Focuses the analysis on completed deliveries.
BR-02	Delivery Delay Days = Actual Delivery Date – Estimated Delivery Date.	Measures delivery performance against the promised date.
BR-03	Delay Days > 0 = Delayed; Delay Days ≤ 0 = On Time.	Creates a consistent delivery-status classification.
BR-04	Exclude delivered orders without an actual customer delivery date from delay calculations.	Prevents incomplete records from affecting delivery KPIs.
BR-05	Use review_score (1–5) as the customer satisfaction measure. If an order has multiple review records, average the scores at order level.	Creates a consistent order-level satisfaction measure.
BR-06	Use customer_state for geographic analysis, linked to orders through customer_id.	Identifies geographic patterns in delivery delays.
BR-07	Customer satisfaction comparisons are performed at order level so an order with multiple review records is not counted multiple times.	Prevents over-weighting orders with multiple review records.

4. Key Analytical Measures

Measure	Definition
Delay Rate	$\text{Delayed delivered orders} \div \text{eligible delivered orders} \times 100.$
On-Time Delivery Rate	$\text{On-time delivered orders} \div \text{eligible delivered orders} \times 100.$
Average Delay Days	Average delay days among eligible delayed delivered orders.
Average Review Score – Delayed	Average order-level review score for delayed orders.
Average Review Score – On Time	Average order-level review score for on-time orders.

5. Data Relationships

- Orders → Customers: customer_id
- Orders → Order Reviews: order_id

6. Analytical Limitation

The analysis identifies an association between delivery delay and customer review scores; it does not establish that delivery delay alone caused lower ratings.